

LOOPS SOLUTIONS

Solution 1: Hello is printed 2 times.

Solution 2:

```
import java.util.Scanner;

public class Solution {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int number;
        int choice;
        int evenSum = 0;
        int oddSum = 0;

        do {

            System.out.print("Enter the number ");
            number = sc.nextInt();

            if( number % 2 == 0) {
                evenSum += number;
            } else {
                oddSum += number;
            }

            System.out.print("Do you want to continue? Press 1 for yes or 0 for no");

            choice = sc.nextInt();

        } while(choice==1);

        System.out.println("Sum of even numbers: " + evenSum);
        System.out.println("Sum of odd numbers: " + oddSum);

    }

}
```

Solution 3:

```
import java.util.Scanner;

public class Solution {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int num; // To hold number
        int fact = 1; // To hold factorial

        System.out.print("Enter any positive integer: ");
        num = sc.nextInt();

        for(int i=1; i<=num; i++) {
            fact *= i;
        }

        System.out.println("Factorial: " + fact);
    }
}
```

Solution 4:

```
import java.util.*;

class MultiplicationTable {

    public static void printMultiplicationTable(int number){
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number:");
        int n = sc.nextInt();
        for(int i=1; i<=10; i++) {
            System.out.println(n + " * " + i + " = " + n*i);
        }
    }

    public static void main(String s[]) {
        printMultiplicationTable(5);
    }
}
```

Solution 5:

Scope of variable is referred to the part of the program where the variable can be used.

In this program variable `i` is declared in the `for` loop. So scope of variable `i` is limited to the `for` loop only that is between `{` and `}` of the `for` loop. There is a `display` statement after the `for` loop where variable `i` is used which means `i` is used out of scope. This leads to compilation errors.

The program given will not run and give an error instead. To correct the program, the variable `i` needs to be declared outside the `for` loop.

